

Model Paper - Database Management Systems

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Q1 (20 marks)

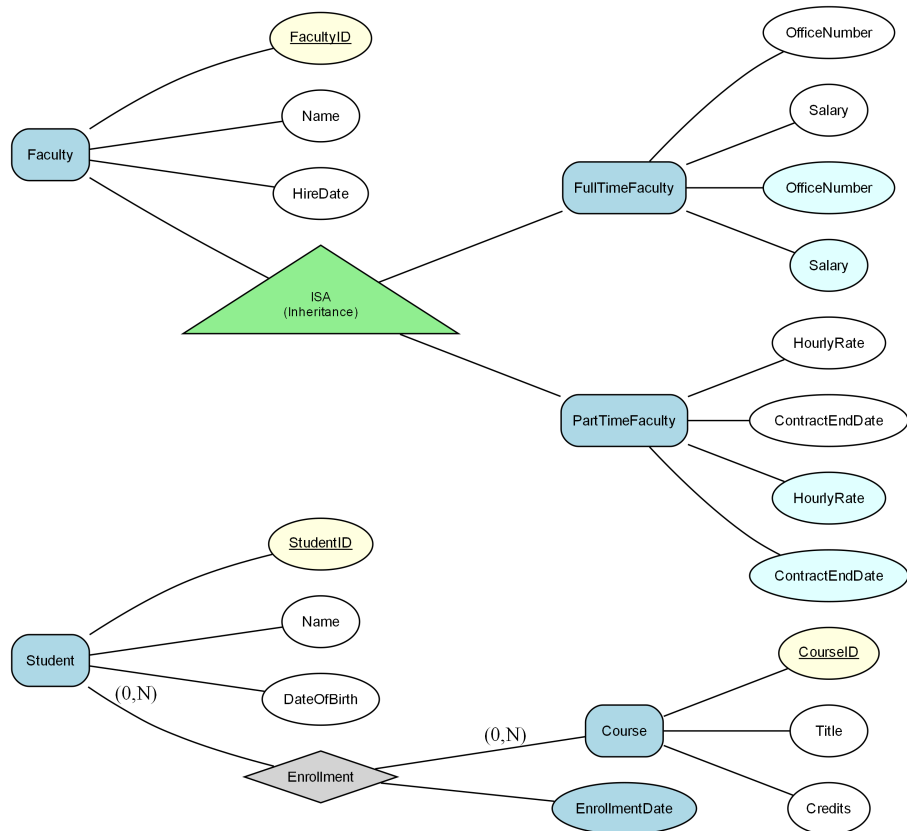
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university database tracks students, courses, and faculty members. The Student entity includes attributes such as StudentID, Name, and DateOfBirth. The Course entity has attributes such as CourseID, Title, Credits. has CourseID, CourseName, and Credits. The Faculty entity has attributes such as FacultyID, Name, Description. consists of FacultyID, FacultyName, and HireDate. Additionally, there is an Enrollment relationship that connects the Student and Course entities, which includes an attribute EnrollmentDate. The university also categorizes Faculty into two subtypes: FullTimeFaculty, with additional attributes OfficeNumber and Salary, and PartTimeFaculty, with attributes HourlyRate and ContractEndDate. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrollment' relationship (Student participation is optional, Course participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, A \rightarrow B, BD \rightarrow C\}$

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A retail company is designing a database to manage its inventory. The inventory includes products with attributes such as `product_id`, `product_name`, `price`, and `stock_quantity`. The company requires integrity constraints to ensure that the data remains consistent and accurate. Moreover, the database should support operations for adding new products, updating stock levels, and deleting discontinued items while adhering to the required rules.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: ````java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery(); ```` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for managing data integrity in the inventory database. (2 marks)

d) Briefly explain how foreign keys contribute to the integrity of the inventory database. (3 marks)

e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.

i. Write a T-SQL statement to create a login for Sarah with windows authentication. (2 marks)

ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)

iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the operational database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the operational databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), dob: date, address: varchar(150)) Appointment (appointmentId: int, patientId: int, appointmentDate: date, status: varchar(50)) Bill (billId: int, appointmentId: int, amount: real, paymentStatus: varchar(50)). The 'Patient' table stores information about patients, including unique patient ID, name, date of birth, and address. The 'Appointment' table stores details about patient appointments, including the appointment ID, patient ID, appointment date, and status of the appointment. The 'Bill' table keeps track of billing details associated with appointments, including the bill ID, appointment ID, total amount billed, and payment status. The 'Bill' table manages relevant information for the database system.

a) Write SQL Queries to perform the following: i. Find the name and address of the patient who has an appointment scheduled on '2023-10-15'.

i. Find the name and address of the patient who has an appointment scheduled on '2023-10-15'. (4 marks)

ii. Find the patient who has the highest total amount compared to all other patients. Find the patientId and name. (6 marks)

iii. Find the appointmentDate, status, and name from Appointment, Patient, and Appointment where Appointment.appointmentDate is not null. (7 marks)

b) Create a function to calculate the total bill amount for a given patient. This function should account for bill with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Bill' table whenever a new bill record is added, or an existing bill record is updated. The 'amount' column should contain the total bill amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total bill amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)